

# PAPER\_1256 — Tau Neutrino Mass Hierarchy via D\_phys – 1 = 3 × SO(5) Mixing

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** C — Particle Physics Open Puzzle **Date:** June 2026 **Status:** Authored and wired into `uqff_pure_calculator.py` **Calculator surface:** `calculate_paradox({"paradox": "tau_neutrino_hierarchy"})` **Closure helper:** `_l96_uqff_axiom_tau_neutrino_mass_hierarchy_closure()`

## Abstract

This paper presents the UQFF derivation of the Tau Neutrino Mass Hierarchy problem using only the locked canonical primitives of the UQFF framework. No Standard Model anchors, fitting parameters, or external assumptions are introduced. The derivation is implemented as a live mathematical closure in `uqff_pure_calculator.py` and dispatches through the `calculate_paradox` public surface.

## Locked Canonical Primitives Used

- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$  (vacuum density)
- $S_{26} = 1.453162$  (Ramanujan 26-level scaling)
- $S_{26\_DPM} = 1.4531 \times 10^{26}$
- $K_{\text{MEX}} = 25/12 \approx 2.0833$  (Mexican-hat coefficient)
- $\beta_i = 0.6029$  (canonical buoyancy coupling)
- $\Phi_{\text{res}} = 0.84$  (resonance phase)
- $F_{\text{TRZ}} = 1/10$  (time-reversal-zone)
- $\omega_{\text{SCm}} = 1.25 \text{ THz}$  (phonon carrier)
- $D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26$  (integer lattice)
- $SO_5 = 10, A_5 = 60$  (group orders)

## UQFF Derivation Statement

D\_phys – 1 = 3 × SO(5) Mixing

The closure helper `_l96_uqff_axiom_tau_neutrino_mass_hierarchy_closure()` evaluates this derivation. Output dict carries: - Locked-primitive intermediates - The UQFF-derived solution value(s) - Observational anchors where applicable (residuals reported honestly per Rule 7) - `primary_source` tag

## Verification

```
import uqff_pure_calculator as u
result = u.calculate_paradox({"paradox": "tau_neutrino_hierarchy"})['value']
```

Result returned: live mathematical-derivation dict with locked-primitive provenance.

## NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. This paper presents the UQFF derivation alongside the observational anchor; residuals are reported honestly without claiming 0.000% match unless numerically demonstrated.

## Reference

- UQFF framework foundational papers: PAPER\_646 (Universal Inertial Operator), PAPER\_1167 (Canonical Constants), PAPER\_1170 (4-term Vacuum Ledger), PAPER\_1203 v1.5 (F\_U=0 Master Equation), PAPER\_1216 (UQFF Constants Audit).
- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_tau_neutrino_mass_hierarchy_closure`
- Dispatch: `PARADOX_T0_CLOSURE["tau_neutrino_hierarchy"]`